

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2275

Orebank 100 kV
January 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2275. The Interconnection Customer proposes a 25.7 MW battery energy storage facility interconnecting at the Orebank 100 kV substation in the Ohio area. The request was received on January 09, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 447 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,262,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kepler-88Th-Fruitridge 100 kV	Geauga-Guayanes 2902 100 kV	130.4	154	6.1
Progressive-Lark 100 kV Line	Kepler-High Bridge 100 kV	125.0	161	8.9
Wethersfield-Progressive 100 kV	Kepler-High Bridge 100 kV line	121.0	201	8.9
High Bridge-Kepler 100 kV	Geauga-Guayanes 2902 100 kV	119.3	195	6.1
Portage Lakes-Weedpatch 230 kV	Weedpatch-Surrowiec 500 kV Line	118.7	875	5.1
Lester-Guayanes 2902 100 kV L	Kepler-High Bridge 100 kV	108.3	144	5.7
Guayanes 2902-Geauga 100 kV	High Bridge-Kepler 100 kV L	108.1	163	5.7
El Paso 500 kV/345 kV TX	Ganso-El Paso 500 kV	104.5	864	5.6
Midway-Siena-Orebank 100 kV #2	Midway-Siena-Orebank 100 kV #1	100.9	166	15.6
Galeville-Marcs 100 kV	Imperial-Eufaula 500 kV L	100.3	377	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kepler-88Th-Fruitridge 100 kV	Rebuild line with higher-capacity conductor	\$1,320,000	100% (\$1,320,000)
Progressive-Lark 100 kV Line	Rebuild line with higher-capacity conductor	\$1,602,000	26% (\$423,000)
Wethersfield-Progressive 100 kV	Rebuild line with higher-capacity conductor	\$2,391,000	26% (\$631,000)
High Bridge-Kepler 100 kV	Rebuild line with higher-capacity conductor	\$1,111,000	100% (\$1,111,000)
Portage Lakes-Weedpatch 230 kV	Rebuild line with higher-capacity conductor	\$28,046,000	1% (\$243,000)
Lester-Guayanes 2902 100 kV L	Reconductor line	\$1,379,000	100% (\$1,379,000)
Guayanes 2902-Geauga 100 kV	Reconductor line	\$751,000	100% (\$751,000)
El Paso 500/345 kV Auto	Replace transformer with higher-rated unit	\$28,410,000	1% (\$194,000)
Orebank-Midway-Siena 100 kV Line	Reconductor line	\$404,000	31% (\$125,000)
Galeville-Marcs 100 kV	Reconductor line	\$1,643,000	5% (\$85,000)
	Total allocated to project		\$6,262,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.